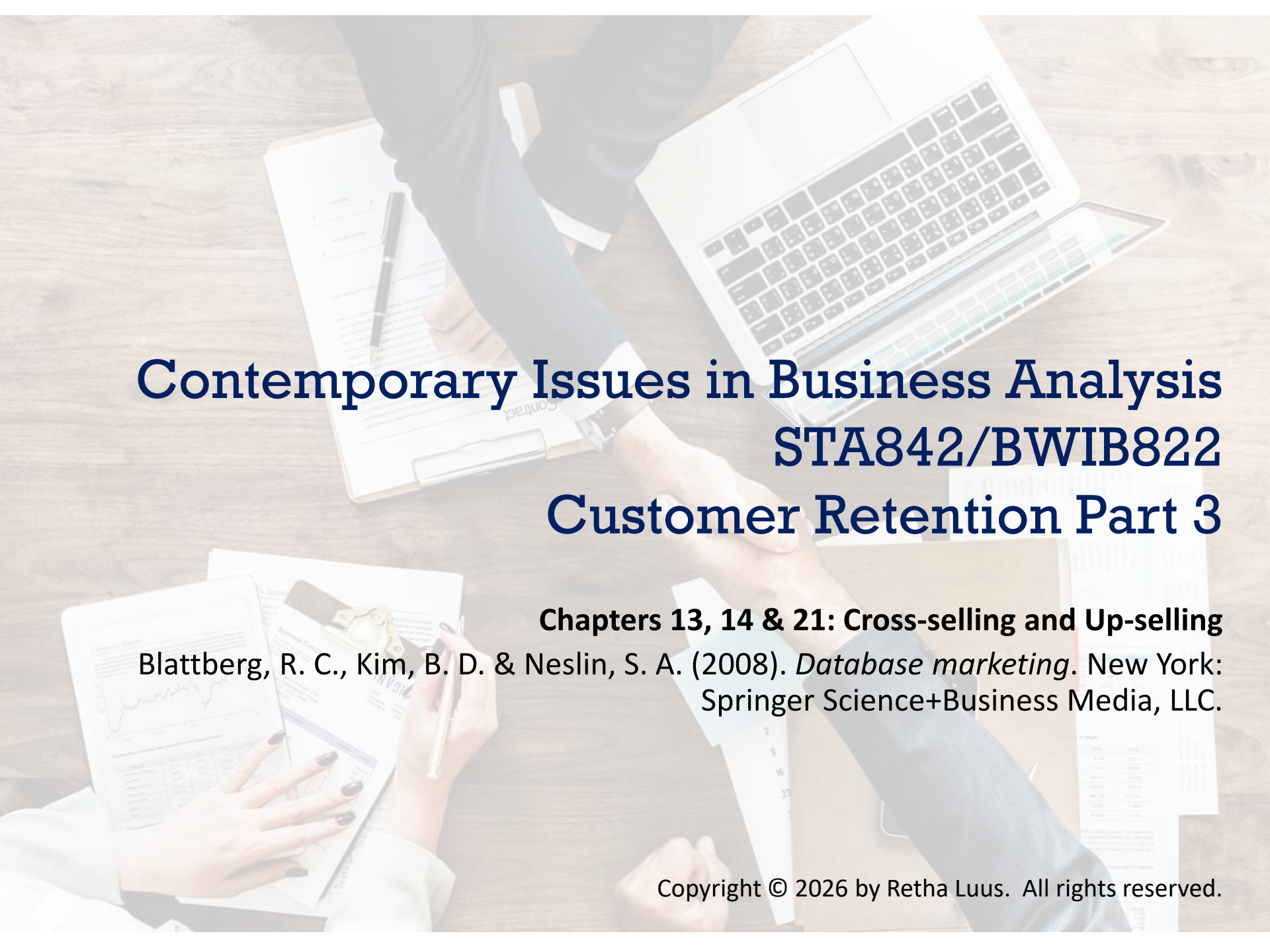




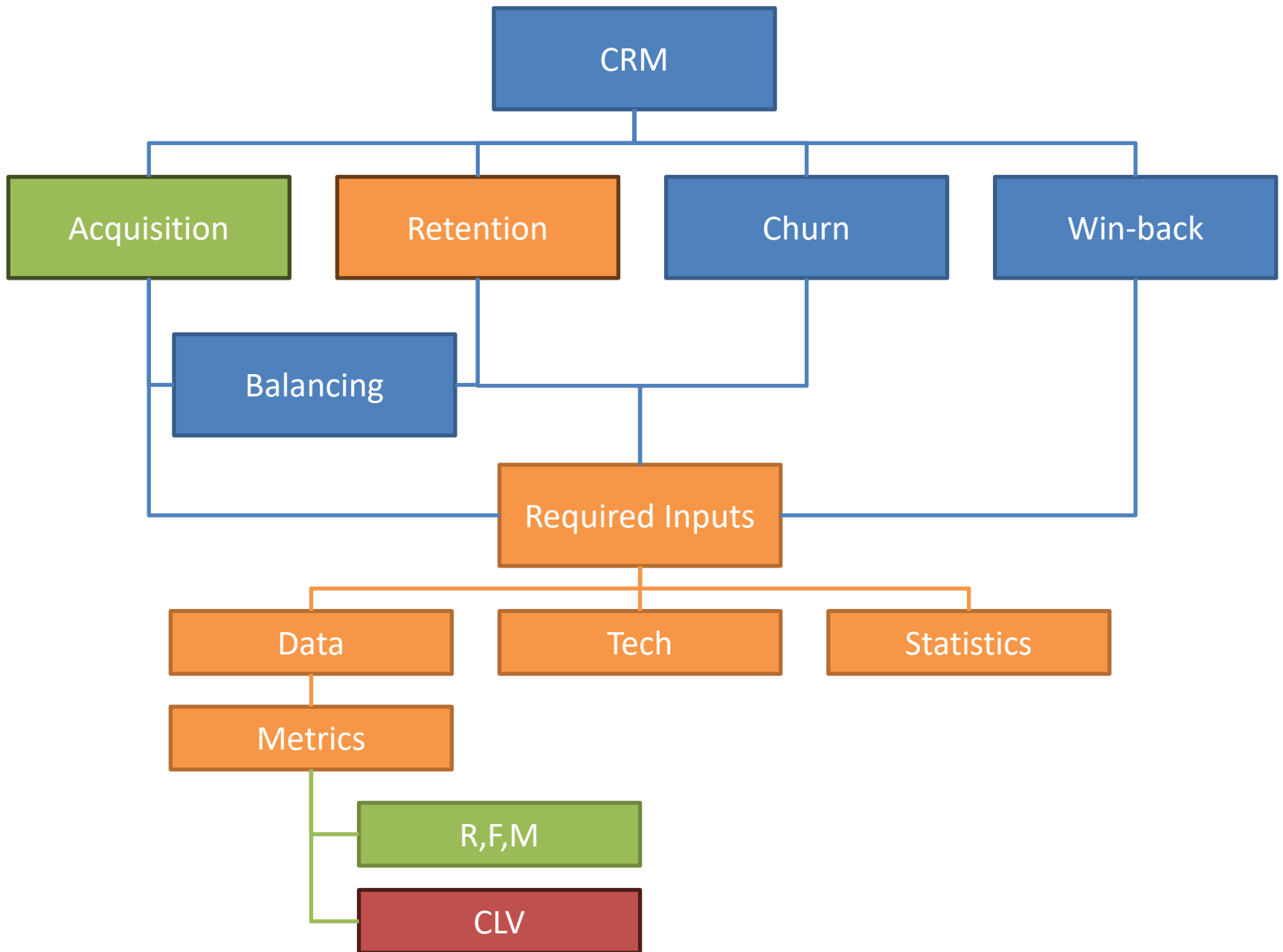
Contemporary Issues in Business Analysis

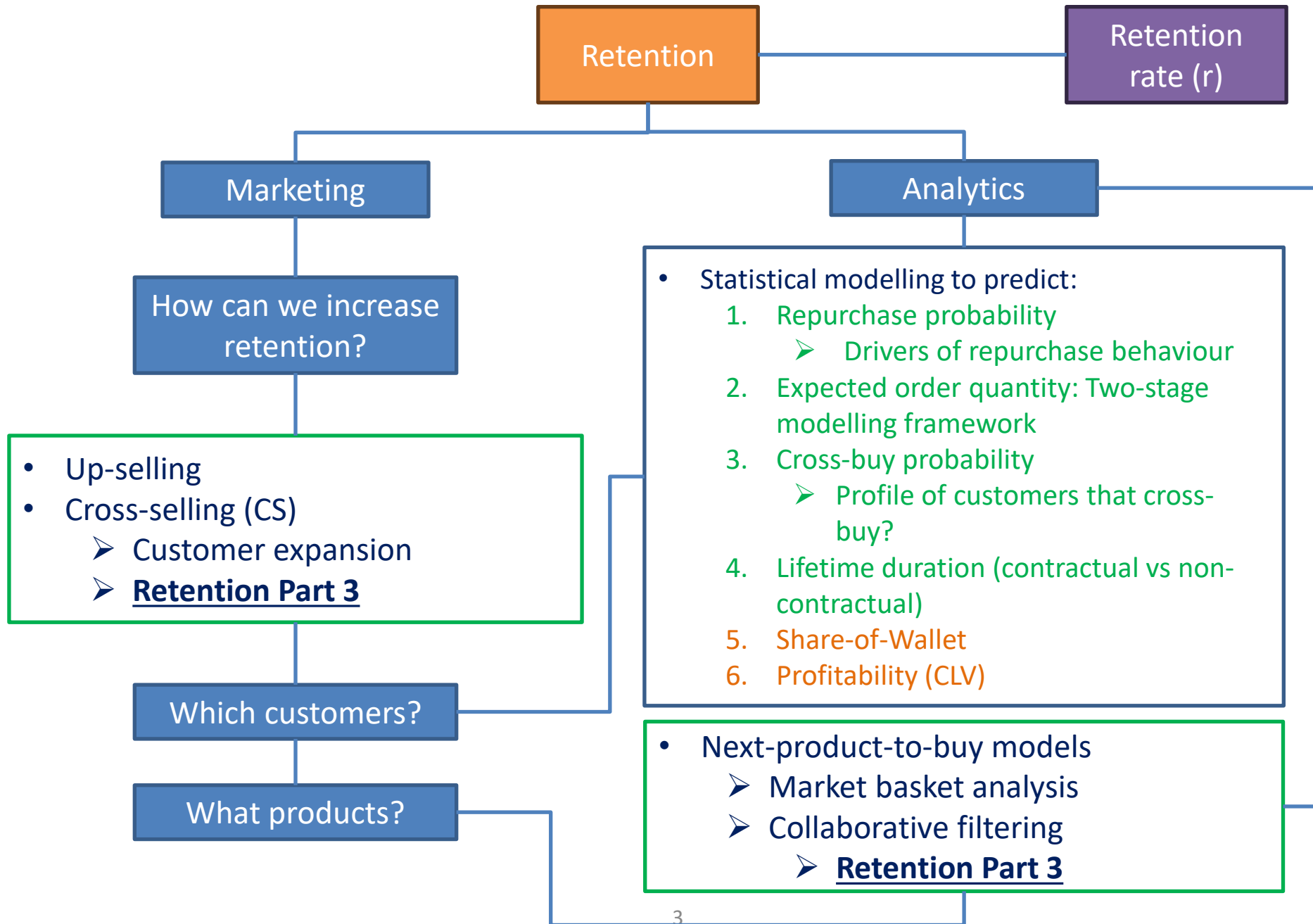
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Customer Retention Part 3

Chapters 13, 14 & 21: Cross-selling and Up-selling

Blattberg, R. C., Kim, B. D. & Neslin, S. A. (2008). *Database marketing*. New York: Springer Science+Business Media, LLC.





Introduction

- **Cross-selling** and **up-selling** are important strategies in **customer retention**
- **Cross-selling**: when the firm **sells different products** to its customers
 - **Ex**: customer owns an iPhone, want to sell customer an iPod
- **Up-selling**: when the firm **sells more of the same product** to the customer
 - **Ex**: customer owns a Dell laptop, offer them a faster laptop

Introduction

- Consider a **simplified retention model of CLV**

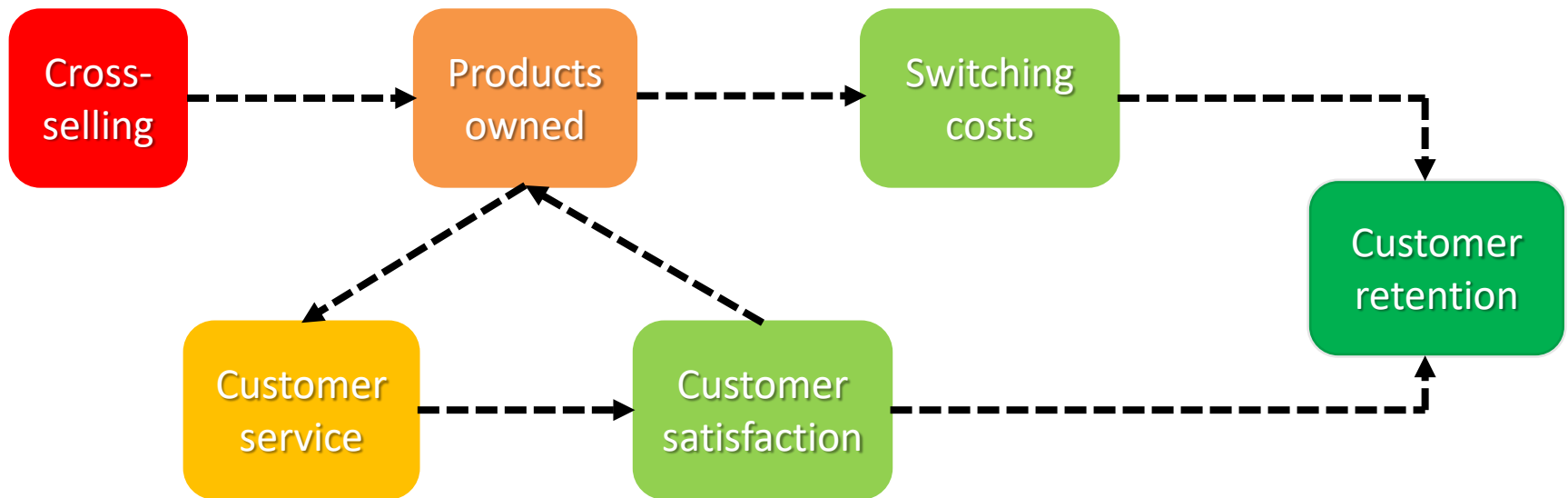
$$CLV = \sum_{t=1}^{\infty} \frac{margin_t \times retention^{t-1}}{(1 + \delta)^{t-1}}$$

- $margin_t$: customer's profit contribution in period t
- $retention$: retention rate
- δ : discount rate

- Three potential **benefits** of cross-/up-selling:
 1. can generate **higher sales** in the **current period**
 - **E.g.**: $t = 1$ customer buys new phone
 2. can **increase future revenue**
 - **E.g.**: $t > 1$ customer adds data bundle to monthly contract
 3. might **increase retention rate** (**Figure 21.1**, **slides 6 - 8**)

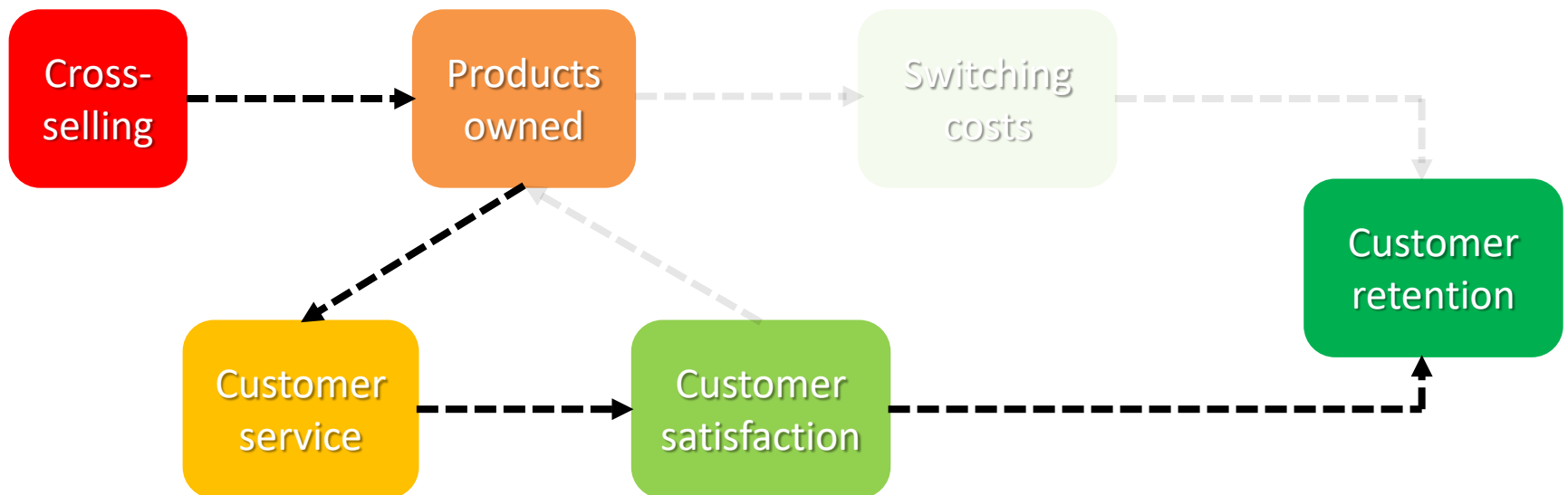
Introduction

Figure 21.1, p. 516, Blattberg et al. (2008)



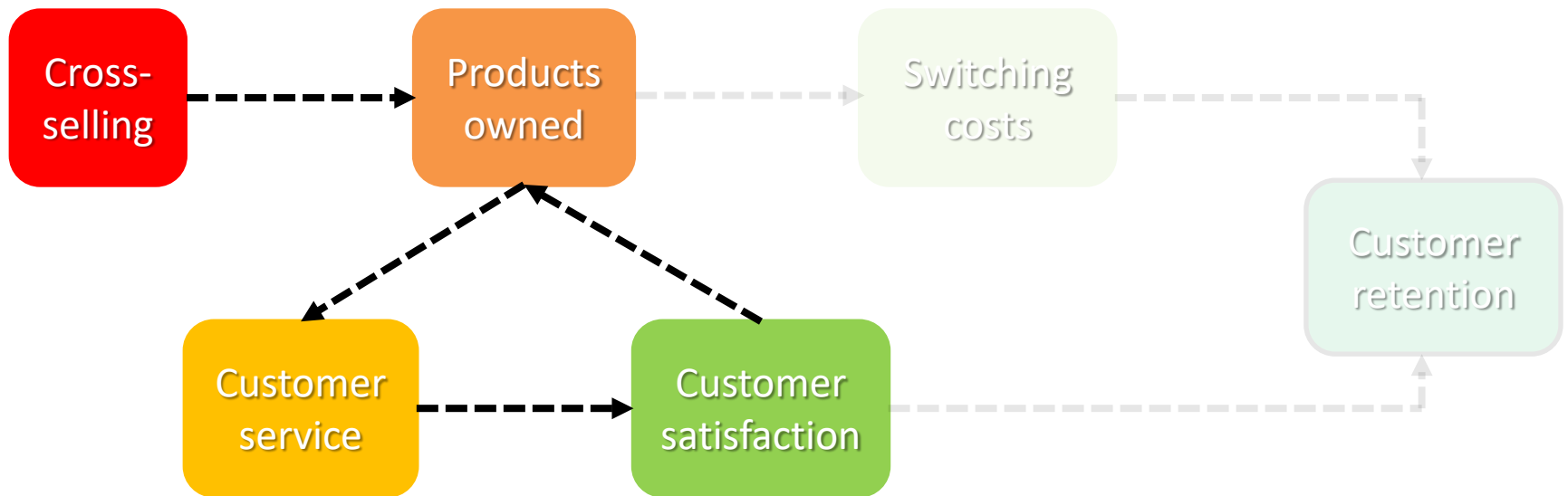
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Figure 21.1, p. 516, Blattberg et al. (2008)



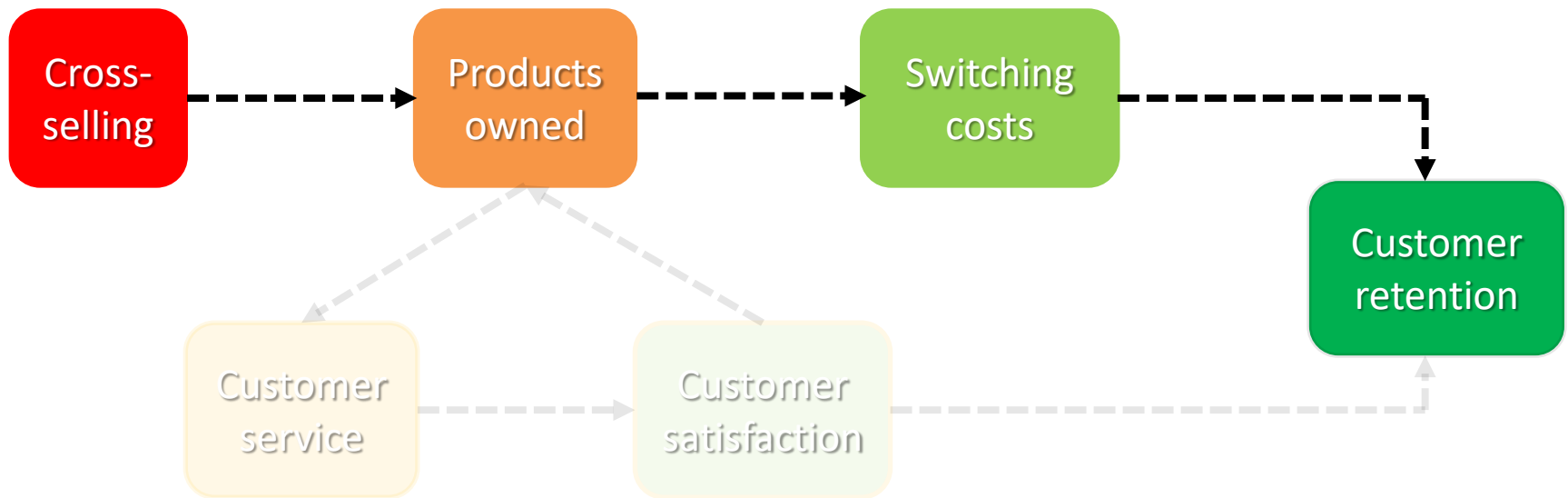
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Figure 21.1, p. 516, Blattberg et al. (2008)



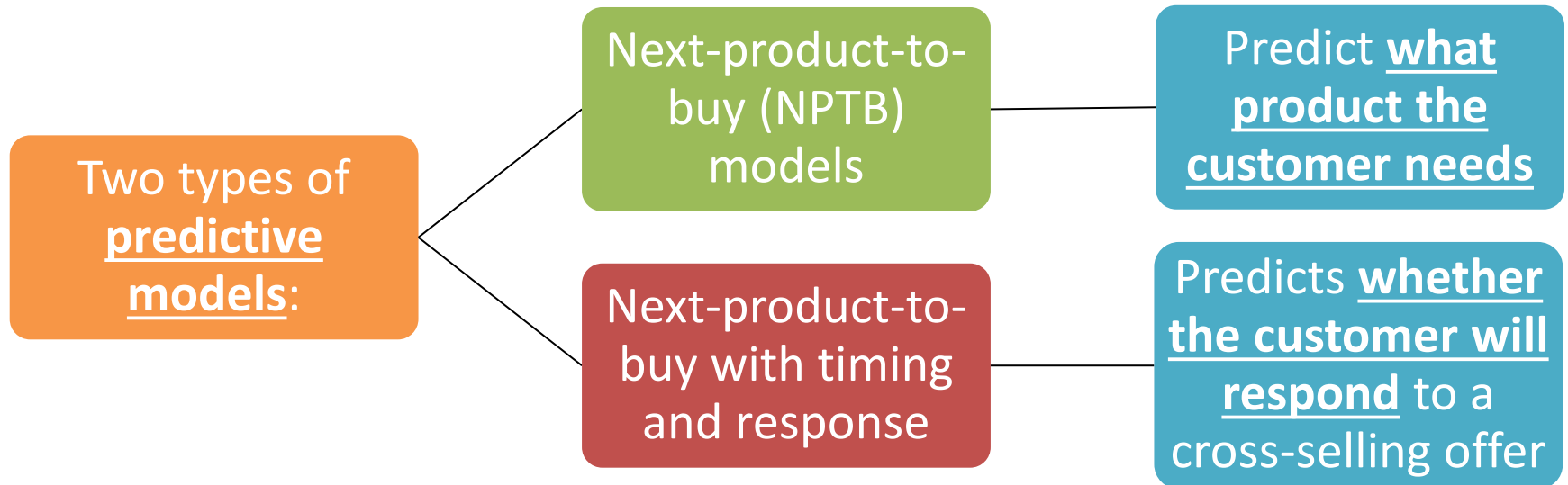
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Figure 21.1, p. 516, Blattberg et al. (2008)



Cross-selling Models

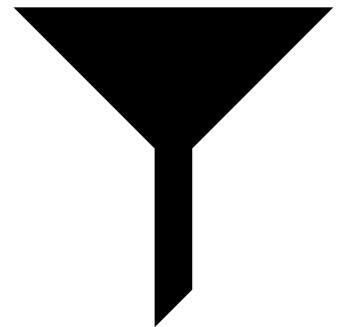
Key question: what products should the firm sell at what time to which customers?



NEXT- PRODUCT-TO- BUY MODELS

Market basket analysis
(Chapter 13, Blattberg et al.)

Collaborative filtering
(Chapter 14, Blattberg et al.)



Next-product-to-buy (NPTB)

- **Strategy**: predict the product the customer is most likely to buy next and then cross-sell that product
- **Ideal**: survey customer and ask what product they need
- **Reality**: look at products customers with similar profiles (current product ownership, personal characteristics, etc.) have bought next
 - Assume this product is what the customer needs
- Most basic **NPTB approaches**:
 - Market basket analysis (MBA)
 - Collaborative filtering

Market basket analysis (MBA)



- **MBA** analyses products customers tend to buy together
- Typical MBA
 - Input is customer **transaction data**
 - **Analyse** all combinations of products bought together
 - Output is a series of product **association rules**
- **Association rules** used to decide **which products to cross-sell** or promote together
- Value obtained from MBA
 1. Association rules can inform how shelves need to be stocked
 2. Design of promotional strategies
 3. **Selection of cross-selling items**

Market basket analysis (MBA)



- **Association rules** consist of an **antecedent** and a **consequent**
- Consider **example** transaction data (Table 13.1, Blattberg et al.)

Transactions	Items purchased (basket)
1	Milk, orange juice, ice cream, beer, soap
2	Milk, ice cream, beer
3	Milk, orange juice, detergent
4	Milk, ice cream, pizza
5	Milk, orange juice, soap

- Intuitively we can identify a number of associations, but we are looking for **interesting, relevant, good** rules
- **Three** most popular **metrics** to evaluate the quality/strength of the association rules: **support**, **confidence** and **lift**

Market basket analysis (MBA)



- **Support** is the % of transactions that contain a specific combination of items
- Indication of **how often the combination** of items **occurs** out of all transactions in the database
- Support for the **association rule** $A \rightarrow B$:

$$\begin{aligned} \text{Support}(A \rightarrow B) &= \frac{\# \text{ baskets that contain } A \text{ and } B}{\text{total \# baskets}} \\ &= P(A \text{ and } B) \\ &= P(B \text{ and } A) \\ &= \text{Support}(B \rightarrow A) \end{aligned}$$

Transactions	Items purchased (basket)
1	Milk, juice, ice cream, beer, soap
2	Milk, ice cream, beer
3	Milk, juice, detergent
4	Milk, ice cream, pizza
5	Milk, juice, soap

$$\text{Support}(\text{Milk} \rightarrow \text{Beer}) = 0.4$$

$$\text{Support}(\text{Milk}) = 1$$

⇒ All transactions include milk, only 40% of those include beer too (**powerful antecedent**).

Disadvantage of Support as evaluation metric

Market basket analysis (MBA)



- **Confidence** measures how much the consequent depends on the antecedent (**strength** of association)
- **Conditional probability** of basket containing item B if it already contains item A
- Confidence for the **association rule** $A \rightarrow B$:

$$\begin{aligned} & \text{Confidence}(A \rightarrow B) \\ &= \frac{\# \text{ baskets that contain } A \text{ and } B}{\# \text{ baskets that contain } A} \\ &= P(B|A) \\ &\neq P(A|B) \\ &\quad \Rightarrow \text{Confidence}(A \rightarrow B) \\ &\quad \neq \text{Confidence}(B \rightarrow A) \end{aligned}$$

Transactions	Items purchased (basket)
1	Milk, juice, ice cream, beer, soap
2	Milk, ice cream, beer
3	Milk, juice, detergent
4	Milk, ice cream, pizza
5	Milk, juice, soap

$$\begin{aligned} & \text{Confidence}(\text{Ice cream} \rightarrow \text{Juice}) \\ &= 0.33 \end{aligned}$$

$$\begin{aligned} & \text{Expected Confidence}(\text{Juice}) \\ &= P(\text{Juice}) = 0.6 \end{aligned}$$

$\Rightarrow$ Randomly selected transaction 60% chance of including juice (**powerful consequent**)

Disadvantage of Confidence as evaluation metric

Market basket analysis (MBA)



- **Lift** – improvement/impact – a measure to **overcome the problems** with support and confidence
- The factor by which confidence exceeds expected confidence
- Lift for the **association rule** $A \rightarrow B$:

$$\begin{aligned} \text{Lift}(A \rightarrow B) &= \frac{\text{Confidence}(A \rightarrow B)}{\text{Expected Confidence}(B)} \\ &= \frac{P(A \text{ and } B) / P(A)}{P(B)} \\ &= \frac{P(A \text{ and } B)}{P(A)P(B)} \\ &= \text{Lift}(B \rightarrow A) \end{aligned}$$

- Larger lift, more interesting association rules

Transactions	Items purchased (basket)
1	Milk, juice, ice cream, beer, soap
2	Milk, ice cream, beer
3	Milk, juice, detergent
4	Milk, ice cream, pizza
5	Milk, juice, soap

$$\text{Lift}(\text{Pizza} \rightarrow \text{Ice cream}) = 1.67$$

$$\text{Support}(\text{Pizza}) = 0.2$$

⇒ “Pizza → Ice cream” looks good, but very few buy pizza (**weak antecedent**).

Disadvantage of Lift as evaluation metric

Market basket analysis (MBA)



Several issues with using MBA for cross-selling

- Disadvantages of support, confidence and lift
- Selection of time period to construct confidence metric
- Which products to consider out of a very large product line
- We looked at one antecedent and one consequent – want to make more recommendations?
 - Collaborative filtering a possible solution

Collaborative filtering models



- A step up from MBA as it takes into account multiple antecedents
- Some examples currently in practice to follow

Collaborative filtering models



The screenshot shows the Takealot.com website interface. At the top, there's a navigation bar with the Takealot logo, a search bar, and links for Help, Sell on Takealot, Login, Register, Orders, and My Account. Below the navigation bar, there's a category filter for 'Shop by Department' and a list of departments: All Gaming, Pre-Orders, Playstation, Xbox, Nintendo, PC Gaming, Gaming Merchandise, and Gaming Accessories. The main content area displays a product page for a 'USB Full HD Web Cam 1080p with Mic'. The product is shown with a large image and a smaller thumbnail. The price is R 249, with a 50% OFF tag. The product is in stock and has 3.8 stars from 187 reviews. To the right of the product, there's a sidebar with the price R 249, a green 'Add to Cart' button, and an 'Add to List' button. Below the product, there's a 'You Might Also Like' section highlighted with a red dashed border. This section contains four product recommendations, each with a thumbnail, title, price, rating, and an 'Add to Cart' button.

Product Name	Price	Rating	Reviews
Volkano Zoom 700 Webcam	R 299	4.1	19
HD-1080P Webcam with Microphone USB	R 232	2.8	20
1080P HD Webcam	R 309	3.8	26
Astrum Full HD USB Webcam With Mic	R 329	4.4	5

Collaborative filtering models



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Details

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Roll over image to zoom in

VIDEO

£129.95

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In stock.

Quantity: 1

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Frequently bought together

 +  Total price: **£258.95**
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☒ Bose 752195-0100 SoundLink Color II: Portable Bluetooth, Wireless Speaker with Microphone - Black, 13.1 cm*5.55 ... £129.00

Products related to this item

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Page 1 of 35

Collaborative filtering models



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Procedia Computer Science
Volume 49, 2015, Pages 136-146

Matrix Factorization Model in Collaborative Filtering Algorithms: A Survey ☆

Dheeraj Bokde, Sheetal Girase, Debajyoti Mukhopadhyay A

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Abstract

Recommendation Systems (RSs) are becoming tools of choice to select the online information relevant to a given user. *Collaborative Filtering* (CF) is the most popular approach to build Recommendation System and has been successfully employed in many applications. Collaborative Filtering algorithms are much explored technique in the field of Data Mining and Information Retrieval. In CF, past user behavior are analyzed in order to establish connections between users and items to recommend an item to a user based on opinions of other users. Those customers, who had similar likings in the past, will have similar likings in the future. In the past decades due to the rapid growth of Internet usage, vast amount of data is generated and it has become a challenge for CF algorithms. So, CF faces issues with sparsity of rating matrix and growing nature of data. These challenges are well taken care of by *Matrix Factorization* (MF). In this paper we are going to discuss different Matrix Factorization models such as Singular Value Decomposition (SVD), Principal Component Analysis (PCA) and Probabilistic Matrix Factorization (PMF). This paper attempts to present a comprehensive survey of MF model like SVD to address the challenges of CF algorithms, which can be served as a roadmap for research and practice in this area.

Part of special issue:
Proceedings of 4th International Conference on Advances in Computing, Communication and Control (ICAC3'15)
Edited by Srija Umrikishan, Sunil Surve, Deepak Shor

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Collaborative filtering models



Very popular in current era of
recommendation engines

- You might also like.....
- Frequently bought together.....
- Recommended articles.....

Two major forms:

- Memory-based / User-based
- Model-based / Item-based

Collaborative filtering models



- **Table 14.1**, p. 354, Blattberg et al.

	MOVIE 1	MOVIE 2	MOVIE 3	MOVIE 4	MOVIE 5
Amy	5	2		4	1
Joseph	1		1	2	
Michael		4	3		5
Jim	3	1		1	2
Laura	5	3	4		1

*Ratings matrix

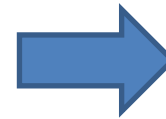
- **Objective**: make movie recommendations to customers **that they might like**
 - **Predict missing ratings** from other observed ratings

Collaborative filtering models



General: n users, m items, U_a the active user, I_p the target item, $r_{i,j}$ rating by user i on item j

RATINGS MATRIX (INPUT)					
	ITEMS				
USERS	I_1	...	I_p	...	I_m
U_1	$r_{1,1}$	...	$r_{1,p}$	...	$r_{1,m}$
$\vdots$	$\vdots$	...	$\vdots$	...	$\vdots$
U_a	$r_{a,1}$	...	???	...	$r_{a,m}$
$\vdots$	$\vdots$	...	$\vdots$	...	$\vdots$
U_n	$r_{n,1}$	...	$r_{n,p}$	...	$r_{n,m}$

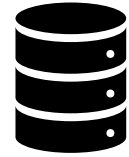


Predict empty cells
(CF algorithm)



Make recommendations
(Top- N list)

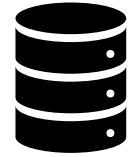
Memory-based



- Also known as user-based or nearest neighbour
- **Predictions** for a target/active user's (U_a) ratings for some target item (I_p) are based on users with similar profiles to the target user
- If these users like the target item, it is recommended to the target user

	Item1	Item2	Item3	Item4	Item5
Alice	5	3	4	4	?
User1	3	1	2	3	3
User2	4	3	4	3	5
User3	3	3	1	5	4
User4	1	5	5	2	1

Memory-based



- For this approach, extract only users that have purchased and rated the target item
- The following equation is used to **predict** the active user's rating of the target item, $r_{a,j}$:

$$\widehat{r_{a,j}} = \bar{r}_a + \frac{\sum_{i=1}^n s_{a,i} (r_{i,j} - \bar{r}_i)}{\sum_{i=1}^n |s_{a,i}|}$$

- Equation consists of **three components**:
 - The active/target user's own average rating: $\bar{r}_a$
 - The observed ratings of target item by other users in the database: $r_{i,j}$
 - Similarity between active/target user and other users that have rated the target item: $s_{a,i}$

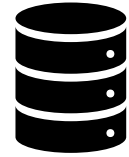
Memory-based



$$\widehat{r_{a,j}} = \bar{r}_a + \frac{\sum_{i=1}^n s_{a,i}(r_{i,j} - \bar{r}_i)}{\sum_{i=1}^n |s_{a,i}|}$$

- Replace missing rating $r_{a,j}$ with active/target user's overall average rating $\bar{r}_a$
- If target/active user doesn't have many ratings, this could severely affect ratings quality
- This won't take account of differences between products/items

Memory-based

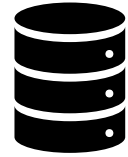


$$\widehat{r}_{a,j} = \bar{r}_a + \frac{\sum_{i=1}^n s_{a,i} (r_{i,j} - \bar{r}_i)}{\sum_{i=1}^n |s_{a,i}|}$$

- Incorporating other users' opinions **could improve** the prediction
- Use **relative preference** for target item j ($r_{i,j}$) compared to overall average rating ($\bar{r}_i$): $(r_{i,j} - \bar{r}_i)$
 - If the user **prefers** item j , this will be a **positive** difference
 - If the user **doesn't prefer** item j , this will be a **negative** difference

However, will the **active/target user also prefer item j (or not)?**

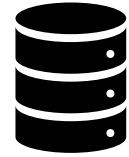
Memory-based



$$\widehat{r_{a,j}} = \bar{r}_a + \frac{\sum_{i=1}^n s_{a,i} (r_{i,j} - \bar{r}_i)}{\sum_{i=1}^n |s_{a,i}|}$$

- This **depends** on whether the active/target user and user i have **similar tastes**
- Use **similarity** to measure this: $s_{a,i}$
 - If they have the **same tastes**, then $s_{a,i}$ will be **positive**
 - If they have the **different tastes**, then $s_{a,i}$ will be **negative**
- $s_{a,i}$ computed over **items rated by both users**

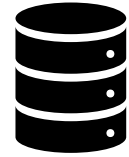
Memory-based



$$\widehat{r_{a,j}} = \bar{r}_a + \frac{\sum_{i=1}^n s_{a,i} (r_{i,j} - \bar{r}_i)}{\sum_{i=1}^n |s_{a,i}|}$$

- Now the relative ratings of users who have rated item j will be combined, but as a **weighted combination**: $s_{a,i} / \sum_{i=1}^n |s_{a,i}|$
 - If user i is **dissimilar** to the active/target user, their relative preference will be **weighted down**
 - If user i is **similar** to the active/target user, their relative preference will be **weighted up**
- This **weighted sum** is then used to **adjust the active/target user's overall average rating** to a more accurate estimate

Memory-based

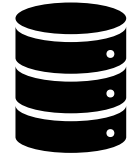


- **Similarity** between target user and other users most important concept
- **Pearson correlation** performs well

$$s_{a,i} = \frac{\sum_j (r_{a,j} - \bar{r}_a)(r_{i,j} - \bar{r}_i)}{\sqrt{\sum_j (r_{a,j} - \bar{r}_a)^2} \sqrt{\sum_j (r_{i,j} - \bar{r}_i)^2}}$$

- Computed over items rated by both users (**unreliability**)
- Using **Pearson correlation** subject to **assumptions** being met
- Consider **Spearman rank correlation** as non-parametric alternative

Memory-based

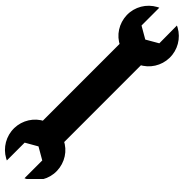


Advantage:

Easy to implement

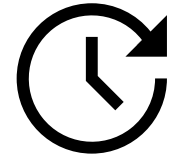
Disadvantages / Limitations:

Data sparsity leads to poor predictive accuracy
Computations grow exponentially (scalability)



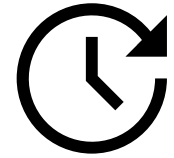
Model-based methods attempt to overcome these limitations

Model-based



- Also known as **item-based** methods
- Considers **set of items** bought and rated by target user
- Two-step process
 1. Calculate **similarity** between items in the set and the target item
 2. Combines similarities into a **predicted preference rating**
- **Two** most popular **similarity measures**:
 1. Pearson correlation coefficient
 2. **Cosine vector**

Model-based



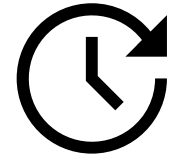
- **Pearson correlation** measure of similarity is calculated as:

$$s_{i,j} = \frac{\sum_{u \in U} (r_{u,i} - \bar{r}_i)(r_{u,j} - \bar{r}_j)}{\sqrt{\sum_j (r_{u,i} - \bar{r}_i)^2} \sqrt{\sum_j (r_{u,j} - \bar{r}_j)^2}}$$

where

- U is the set of all users that **rated both items** i and j
- $r_{u,i}$ is user u 's rating of item i
- $\bar{r}_i$ is the average rating of item i by users in U

Model-based



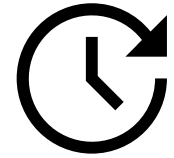
- **Cosine** measure of similarity is calculated as:

$$s_{i,j} = \cos(r_{\bullet i}, r_{\bullet j}) = \frac{r_{\bullet i}^T r_{\bullet j}}{\|r_{\bullet i}\| \|r_{\bullet j}\|}$$

where

- $r_{\bullet i}$ and $r_{\bullet j}$ are columns i and j of the ratings matrix
- $\|r_{\bullet i}\| = \sqrt{\sum_{u=1}^U r_{u,i}^2}$: norm of column i
- Only users that have **rated both items** are included in the similarity calculation

Model-based

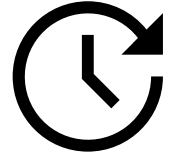


- For this approach, extract only items (I) purchased and rated by the target user
- The following equation is used to predict the rating of item j by the target user u , $r_{u,j}$:

$$\widehat{r_{u,j}} = \boxed{\bar{r}_u} + \frac{\sum_{i=1}^I s_{i,j} \boxed{r_{u,i} - \bar{r}_u}}{\sum_{i=1}^I |s_{i,j}|}$$

- **Note:**
 - $\bar{r}_u$: target user u 's average rating across all I products they've rated
 - $r_{u,i}$: target user's rating of item i
 - $s_{i,j}$: similarity between item i (rated by target user) and item j (target item)

Model-based



Advantages:

Better quality predictions
Fast online recommendations

Disadvantages / Limitations:

Expensive model building
Predictive performance vs scalability



Hybrid collaborative filtering tries to overcome these limitations

Summary of collaborative filtering methods

Bokde, D., Girase, S. & Mukhopadhyay, D. (2015). Matrix factorization model in collaborative filtering algorithms: a survey. *Procedia Computer Science*, 49, 136-146

Collaborative Filtering Technique	Representative Algorithm	Advantages	Limitations
Memory-Based Collaborative Filtering (Neighborhood Based)	• User-Based CF • Item-Based CF	• Easy implementation • New data can be added easily and incrementally • Need not consider the content of items being recommended • Scales well with correlated items	• Are dependent on human ratings • Cold start problem for new user and new item • Sparsity problem of rating matrix • Limited scalability for large datasets
Model-Based Collaborative Filtering	• Slope-One CF • Dimensionality Reduction (Matrix Factorization) Eg. SVD, PCA	• Better addresses the sparsity and scalability problem • Improve prediction performance	• Expensive model building • Trade-off between the prediction performance and scalability • Loss of information in dimensionality reduction technique (SVD)
Hybrid Collaborative Filtering	• Combination of Memory-Based and Model-Based CF	• Overcome limitations of CF such as sparsity and grey sheep • Improve prediction performance	• Increased complexity and expense for implementation

Summary

Market basket analysis ($A \rightarrow B$)

1. Set of association rules (1 antecedent (A), 1 consequent (B))
2. Evaluation of the quality of the association rules using three metrics:
 1. Support: $P(A \text{ and } B)$
 2. Confidence: $P(B|A)$
 3. Lift: $\frac{\text{Confidence}(A \rightarrow B)}{E[\text{Confidence}(B)]} = \frac{P(B|A)}{P(B)}$

Collaborative filtering

- Recommendations
- Ratings matrix, users by items
- Use collaborative filtering to fill in all the missing item ratings
- Memory/user-based
 - Uses similarity between users to predict an item rating
- Model/item-based
 - Uses similarity between items to predict an item rating

Homework



Classwork (in groups)

Retention Part 2 (AFT model)

Due: 21 April at 1pm



Prep for 5 May

Blattberg et al. Chapter 24

Kumar & Petersen Chapter 6

Updated Schedule

- The schedule **post-assessment** week has been updated:
 - There is no BWIN817/COF827 on 12 May $\Rightarrow$ 2nd class test starts at 11h.
 - There is no BWIB818/STA841 on 14 May $\Rightarrow$ CLV lecture etc. at 11h – 14h15.

Week	Lecture date	Topic	Theory preparation		Practical application in SAS	Classwork in groups	Due
			Kumar & Petersen (2012)	Blattberg et al. (2008)			
02-Feb-26	03-Feb-26	Introduction to CRM	Ch 1 - 2	Ch 1 - 4*			
09-Feb-26	10-Feb-26	RFM analysis		Ch 12	RFM analysis part 1	*	
16-Feb-26	17-Feb-26	Customer acquisition part 1		Ch 20	RFM analysis part 2	*	
23-Feb-26	24-Feb-26	TEST 1					
02-Mar-26	03-Mar-26	Customer acquisition part 2	Ch 3.1 to 3.3		Logistic regression + Two-stage modelling	*	
09-Mar-26	10-Mar-26	Customer acquisition part 3	Ch 3.4				
16-Mar-26	17-Mar-26	Customer retention part 1	Ch 4.1, 4.2, 4.4 and 4.5		Survival curves and Cox PH modelling	*	24/3 @ 13h
23-Mar-26	24-Mar-26	Customer retention part 2	Ch 4.3, 4.6 to 4.8		Two-stage modelling revisited	Given - not for submission	
30-Mar-26	31-Mar-26	ASSESSMENT WEEK (30/3 - 13/4; Theory + Practical)					
06-Apr-26	07-Apr-26						
13-Apr-26	14-Apr-26	No class - UWC graduation			AFT modelling	*	21/4 @ 13h
20-Apr-26	21-Apr-26	Customer retention part 3		Ch 21		No practical	
27-Apr-26	28-Apr-26	RECESS (No class)					
04-May-26	05-May-26	Customer churn	Ch 6	Ch 24	Churn prediction	*	12/5 @ 13h
11-May-26	12-May-26	TEST 2 (11H - 12H15)					
	14-May-26	CLV + Admin (11h - 14h15)		Ch 5		No practical	
18-May-26	19-May-26	NO CLASS					
25-May-26	26-May-26	FINAL ASSESSMENTS (25/5 - 12/6; Theory and Practical)					